Unit 6 Test Study Guide
(Exponents & Exponential Functions)

Name: ____________ Date: ____________ Per: ______

Part 1

Topic 1: Exponent Rules

PRODUCT RULE	POWER RULE	QUOTIENT RULE	NEGATIVE EXPONENT RULE
$x^a \cdot x^b = x^{a+b}$	$(x^a)^b = x^{ab}$	$\frac{x^a}{x^b} = x^{a-b}$	$x^{-a} = \frac{1}{x^a}$

WHAT ABOUT ADDING AND SUBTRACTING MONOMIALS?

Combine like terms; exponents do NOT change

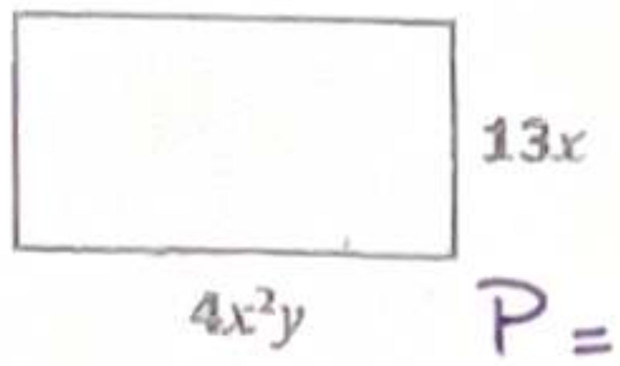
Topic 2: Simplifying Monomials

Simplify each expression completely. Final answer should contain only positive exponents.

1. $6ab - 8ab$ $\boxed{-2ab}$	2. $-2xy^2 - 4xy + 6xy^2$ $\boxed{4xy^2 - 4xy}$	3. Subtract $-6b^5$ from $8b^5$. $8b^5 + +6b^5$ $\boxed{14b^5}$
4. $-7n^{-4} \cdot 5n^{-2}$ $-35n^{-6} = \boxed{\frac{-35}{n^6}}$	5. $(5v^4)^2 \cdot 2v^3 \cdot v$ $25v^8 \cdot 2v^4 =$ $\boxed{50v^{12}}$	6. $(-a^6b)^2 + 9a^{12}b^2$ $a^{12}b^2 + 9a^{12}b^2$ $\boxed{10a^{12}b^2}$
7. $(-2y^4) \cdot (xy^3)^2 - 13x^2y^{10}$ $-2y^4 \cdot x^2y^6$ $-2x^2y^{10} - 13x^2y^{10}$ $\boxed{-15x^2y^{10}}$	8. $\frac{r^6s^7t^2}{r^5s^4t^2}$ $\boxed{rs^3}$	9. $\frac{(-3x^6)^2}{5x^3 \cdot 3x^3}$ $\frac{9x^{12}}{15x^6}$ $\boxed{\frac{3x^6}{5}}$
10. $\left(\frac{4x^4y^2}{6xy}\right)^2$ $\frac{16x^8y^4}{36x^2y^2}$ $\boxed{\frac{4x^6y^2}{9}}$	11. $\frac{7k^{-8} \cdot 3k^{-2}}{6k^2}$ $\frac{21k^{-10}}{6k^2}$ $\boxed{\frac{7}{2k^{12}}}$	12. $\frac{-9n^8}{27n^{10}}$ $\boxed{-\frac{1}{3n^2}}$
13. $\frac{a^{12}b^{-3}}{(ab)^{-4}}$ $\frac{a^{12}b^{-3}}{a^{-4}b^{-4}}$ $\boxed{a^{16}b}$	14. $\frac{3}{4}m^7n^6 \cdot m^5n^9 - \left(\frac{1}{2}m^4n^5\right)^3$ $\frac{3}{4}m^{12}n^{15} - \frac{1}{8}m^{12}n^{15}$ $\boxed{\frac{5}{8}m^{12}n^{15}}$	15. $(-4y^3)^2 \cdot (xy^4)^3 + 7x^3y^{18}$ $16y^6 \cdot x^3y^{12}$ $16x^3y^{18} + 7x^3y^{18}$ $\boxed{23x^3y^{18}}$

Topic 3: Geometric Applications

16. Find the **perimeter** and **area** of the following:



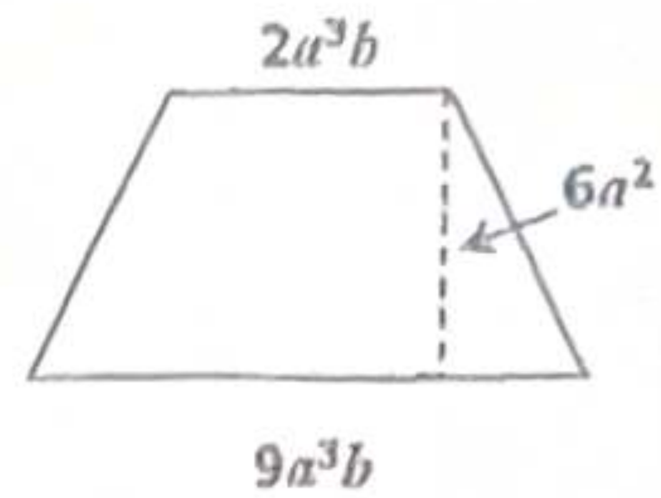
$A = 13x(4x^2y)$

$\boxed{52x^3y}$

$P = 2(13x) + 2(4x^2y)$

$\boxed{8x^2y + 26x}$

17. Find the **area** of the following:



$\frac{1}{2}(2a^3b + 9a^3b)(6a^2)$

$(11a^3b)(3a^2)$

$\boxed{33a^5b}$

Topic 8: Simplifying Radicals

Simplify each expression.

18. $\sqrt{64} + \sqrt[3]{8}$

$8 + 2 = \boxed{10}$

19. $\sqrt[3]{-216} - \sqrt{169}$

$-6 - 13 = \boxed{-19}$

20. $\sqrt{54}$

$\sqrt{9}\ \ 6$

$\boxed{3\sqrt{6}}$

21. $\sqrt{240}$

$\sqrt{16}\ \ 15$

$\boxed{4\sqrt{15}}$

21. $\sqrt{160}$

$\sqrt{16}\ \ 10$

$\boxed{4\sqrt{10}}$

22. $\sqrt{245}$

$\sqrt{49}\ \ 5$

$\boxed{7\sqrt{5}}$

23. $\sqrt{28x^{25}y^{8}}$

$4\ \ 7$

$\boxed{2x^{12}y^4\sqrt{7x}}$

24. $\sqrt{200m^4n}$

$100\ \ 2$

$\boxed{10m^2\sqrt{2n}}$

25. $\sqrt{153a^9b^{10}c^3}$

$\sqrt{9}\ \ 17$

$3a^4b^5c\sqrt{17ac}$

26. $\sqrt{361x^9y^{16}}$

$\boxed{19x^4y^8\sqrt{x}}$

27. $\sqrt[3]{80}$

$8\ \ 10$

$\boxed{2\sqrt[3]{10}}$

28. $\sqrt[3]{-500}$

$-125\ \ 4$

$\boxed{-5\sqrt[3]{4}}$

29. $\sqrt[3]{384}$

$64\ \ 6$

$\boxed{4\sqrt[3]{6}}$

30. $\sqrt[3]{648}$

$216\ \ 3$

$\boxed{6\sqrt[3]{3}}$

Evaluate the expression for numbers 31 – 36.

Solve the equation for numbers 37 – 42.

For numbers 43 – 48, fill in the missing radicand.

31. $\sqrt[3]{8}$ $\boxed{2}$	**32.** $\sqrt[5]{-243}$ $\boxed{-3}$	**33.** $625^{\frac{3}{4}}$ $(\sqrt[4]{625})^{3} =$ $\boxed{125}$
34. $(-25)^{\frac{1}{2}}$ $\sqrt[2]{-25}$ $\boxed{\emptyset}$	**35.** $(\sqrt[5]{8})^4$ $\boxed{5.278}$	**36.** $32^{\frac{3}{5}}$ $(\sqrt[5]{32})^{3}$ $\boxed{8}$
37. $5^x = 5^{3x-2}$ $x = 3x-2$ $-2x = -2$ $\boxed{x=1}$	**38.** $3^{x-1} = 81$ $3^{x-1} = 3^4$ $x-1 = 4$ $\boxed{x=5}$	**39.** $4^{3x} = 8^{x+1}$ $(2^2)^{3x} = (2^3)^{x+1}$ $6x = 3x+3$ $3x \pm 3$ $\boxed{x=1}$
40. $(\frac{1}{2})^{2x+3} = 16$ $(2^{-1})^{2x+3} = 2^4$ $-2x-3 = 4$ $-2x = 7$ $\boxed{x = -7/2}$	**41.** $(\frac{1}{16})^{3x} = 64^{2(x+8)}$ $(16^{-1})^{3x} = (4^3)^{2x+16}$ $(4^2)^{-3x} =$ $-6x = 6x+48$ $-12x = 48$ $\boxed{x=-4}$	**42.** $27^{2x+2} = 81^{x+4}$ $(3^3)^{2x+2} = (3^4)^{x+4}$ $6x+6 = 4x+16$ $2x = 10$ $\boxed{x=5}$
43. $\sqrt{\_\_\_\_} = 2\sqrt{6}$ $\sqrt{4}\sqrt{6}$ $\boxed{\sqrt{24}}$	**44.** $\sqrt{\_\_\_\_} = 3\sqrt{10}$ $\sqrt{9}\sqrt{10}$ $\boxed{\sqrt{90}}$	**45.** $\sqrt{\_\_\_\_} = 5\sqrt{7}$ $\sqrt{25}\sqrt{7}$ $\boxed{\sqrt{175}}$
46. $\sqrt[3]{\_\_\_\_} = 3\sqrt[3]{6}$ $\sqrt[3]{27}\sqrt[3]{6}$ $\boxed{\sqrt[3]{162}}$	**47.** $\sqrt[3]{\_\_\_\_} = 6\sqrt[3]{2}$ $\sqrt[3]{216}\sqrt[3]{2}$ $\boxed{\sqrt[3]{432}}$	**48.** $\sqrt[3]{\_\_\_\_} = -2\sqrt[3]{2}$ $\sqrt[3]{-8}\sqrt[3]{2}$ $\boxed{\sqrt[3]{-16}}$